

Prolegomena to a Critique of Cybernetic Reason

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Abstract—Norbert Wiener’s neologism of Cybernetics, meaning “navigation” or “steering”, immediately resonates with the Kantian period of the three Critiques (1781-1790) book title question *What does it mean to orient oneself in thinking?* (1786). Besides, cognition (*cognitio*; *erkenntnis*) and knowledge (*scientia*; *wissen*) are pivotal concepts in regulatory systems and self-organization. Indeed, the regulative principle of teleology in the “Dialectic of Teleological Judgement” in the *Critique of Judgment* (1790) theoretically expanded but forwent the mechanistic view of nature, next scientifically redeemed by Darwin, nevertheless foreshadowing feedback teleology under subjective autonomy and control. Control theory is rooted on *A Dynamical Theory of the Electromagnetic Field* (1865) by J. C. Maxwell, with electric and magnetic wave synthesis, thus predicting radiation energy pervasiveness in physics. Yet and before, it was Young’s double-slit interference experiment (1801), just about the time of Kant’s death (1804) and prior commissioning of the *Opus Postumum* (ca. 1800) that, while asserting the wave theory of light in anticipation to new interpretations of quantum mechanics, firstly dismantled the *Aufklärung* edifice and architectonics. It is here suggested that not only the background of natural selection and new physics inflicted a fatal blow to transcendental idealism, but also and decisively, that a *coup de grâce* came upon it by a series of diagonalisation arguments ever since Cantor (1891): the negative limits and aftermath results in logic (Frege, Gödel, and Tarski), computation and information (Turing, Shannon) and quantum physics (Heisenberg and Bohr). Cybernetics as a science shall arise as an inversion of Kantian architectonics.

Index Terms—Cybernetics, Kantianism, Architectonics.

I. INTRODUCTION

This paper is intended to settle a philoback and philoforward research program, following the lines of Kantian critical philosophy, to the end of a metaphysical inversion of its classical architectonics (“Critique of Critique”, beyond von Foerster’s *Understanding Understanding*). The method should be the proper transcendental doctrine of the faculty of judgement or analytic of (cybernetic) principles, in relation to a transcendental (cybernated, dynamical and ergodic) dialectic, thus on the foreground to a regulative (regulatory system) principle of reason, anticipating a (complex systems feedback & feedforward) transcendental doctrine of method. Complementary to this is the revision of the period of the

three Critiques (1781-1790). In correspondence with the three Critiques strictly considered, the other triadic paramount texts are “*Prolegomena to any Future Metaphysics that will be able to present itself as a Science*” (1783), “*Answering the Question: What is Enlightenment?*” (1784), and “*What does it mean to orient oneself in thinking?*” (1786), helping to settle cybernetics as a wave-like synthesis future oriented critical science, in articulation with seven appointed prefatory moments of research.

They are: 1) Kantian’s notion of conformity to a natural end (*zweckmässigkeit*) withstanding a prefiguration of Darwin’s natural selection, thus a latent contradiction in terms of scientific validation; 2) the proper metaphysics and synthesis enshrouded in Maxwell’s electromagnetism (1865), which led to the demise of the Newtonian Enlightened corpuscular thing-in-itself (*noumenon*; *ding an sich*); 3) Young’s double-slit experiment (1801), a hallmark in experimental physics that permitted the vindication of the wave-particle duality with radical displacing of the archetype (*archetypon*; *urbild*) of the idea of reason; 4) Cantor’s diagonalisation argument as a fine art constructive proof technique, with profound consequences to the equipotent, leveled, antinomies, wherefrom; 5) formal logic’s upturn, drawing apart from transcendental logic, imploded synthetic *a priori* judgments, hence mathematics’ presumed completeness and consistency transcendental knowledge; 6) the Turing-machine model, which has upheaved a pure associative Humean affinity (*affinität*; *verwandschaft*) practicality; 7) the Copenhagen interpretation (1925-27), with causality and uncertainty postulating critical transcendental empiricism.

II. FROM HUME TO MAXWELL: THE WEALTH OF SCOTTISH ENLIGHTENMENT ASSOCIATES WITH AMPÈRE’S KYBERNETIKOS

In Norbert Wiener’s terms, it is well perceived how the transition from purely mechanistic to electromagnetic automata and servomechanisms ranging several degrees of freedom, was prepared by the the Age of Enlightenment (1715-1789). However, it is strikingly compliant that one such passage is reflected in the greatest herald himself of the philosophy, and author of the reiterative frontispiece adagio of the intellectual, civil and scientific movement: *Sapere Aude*. For, as Immanuel

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Kant agrees to retire from his professorship and the Academic Senate (November, 1801), later dying in Königsberg after a prolonged illness (February, 1804). Alessandro Volta (1779-1804) creates in 1800 the first electric battery, using silver and zinc, alluding to different metal electric potentials, shortly before the English chemist William Nicholson (1753-1815) uses electricity, by the process ever since Faraday known as "electrolysis", to set apart from water molecules, *atoms* of hydrogen and water.

In furtherance, it could almost be said it constitutes an elliptic serendipity that, at the moment Kant finally delivers the *Critique of the Power of Judgment* (April, 1790), thus exploring "bodily" aesthetic judicial and reflective judgments (on the agreeable, the beautiful, the sublime, and the good), the Galvanis are credited to have discovered bioelectricity (1780-1790). But much more conspicuously, at the moment Kant retires from professorship, and the first ever "steering" controlled use of electricity is pioneered, the wave theory of light is experimentally realized. Indeed, in the year of 1801, in the twilight of one of the greatest centuries of western philosophy, two scientific endorsements are, at hindsight, sibylline of the fate of science.

They are French biologist Jean-Baptiste de Lamarck (1744-1829) publication of *Système des animaux sans vertèbres*, where is made the suggestion that organisms change over time due to changes in the environment, in truth, a similarity with Erasmus-to-Charles Darwin pangenesis theory, and a first minor, yet powerful, mover towards the modern "synthetic theory of evolution" (Julian Huxley); and Thomas Young (1772-1829) double-slit experiment, dismissive of Newton's theory of the corpuscular nature of light, thus suggesting the wave theory of light with its inherent physical phenomena – interference, polarization, and diffraction – at the closing of the *Siècle des Lumières*, roughly one century before wave-particle duality becoming the scientific norm in quantum mechanics. In actual fact, if history rightly temporizes the *Siècle des Lumières* with France's bulletin, from the obituary of Louis XIV of France, until the turmoil of the French Revolution (1715-1789), in terms of philosophy of science and cybernated "orientation" in thinking, it is fairer to say that Enlightenment persisted ever since Isaac Newton began to develop work on finite quantities, or "fluxions", and works on prisms separating light into its component and rainbow colors (1665-66), preparing the vernacular publication of *Opticks* (1704), until, precisely, Thomas Young double-slit experiment (1801). Ensuing Thomas Young interference experiment, the British polymath himself proposed the understanding of transverse, rather than acoustic-influenced longitudinal, wave lights, which paved the way for the lightning fast-moving work of Augustin Jean Fresnel (1788-1827).

Remembering that the main obstacle to the wave theory of light was its rectilinear propagation, Fresnel first operated the *synthesis* of Huygen's (1629-1695) secondary waves, with isotropic interference patterns in homogeneous media, which implied diffraction by straight edges, and sinusoidal color patterns. As physicists worked in optics by analysing refraction in

prisms and reflection in mirrors, Étienne-Louis Malus (1775-1812) was able to discover polarized light through double refraction, followed by reflection, in crystals.

It was, then, the opportunity for Fresnel to explain the refraction of "two poles" by way of vertical and horizontal components (thereupon acknowledged as orthogonal vibrations of the "aether", *diagonal* to the ray direction), which implied *pure* light transverse behaviour. Needless to say, this almost necessarily antinomical and contradictory, wave-like status of light, with two-sidedness oscillations under the total amplitude, would provoke the collapse of the corpuscular longitudinal, rectilinear and moral *practical* ascending, neo-Newtonian (notwithstanding the presence of wave-theoretic elements) program of Laplace and Kant.

In fact, certainly not in the case of Laplace's materialistic determinism, but inarguably in the case of Kant, it could be contended that one such moral *practical* ascending was hagiographic, not to say, permitted by its "transcendental liberalness" [1] (A. T. Nuyen, 2002), one of the kind of a miraculous metaphysical cosmology.

In other words, very rapidly, even before the turn of the 19th century, there were virtually no remaining defendants of the long-lived philosophical atomist "corpuscles" (1637) set forth by Descartes (1696-1650). The great irony was that the wave theory of light would itself, also very rapidly, give rise to the particle, or atomic theory (Dalton-Avogadro-Brown-Einstein). There are mainly two debates surrounding the ascension from corpuscular to mediumist concepts of light, interference having been the first, and second to this polarization by action of reflection, or else refraction, through optimally transparent media. The companion Young & Fresnel, on the premise of the work of both Huygens (1629-1695) and, to a lesser extent Euler (1707-1783), in an inherent diagrammatic and geometrical style (a Greek imprint that remained in place until Newton) *vs.* the algebraic tradition of Euler, Lagrange, and Laplace, also assist in understanding, not only the *calculus* newly sanctified pedestal of synthetic *a priori* judgements, but also the originality of the interference experiment, and the whole ensemble of Young & Fresnel collaboration.

A faint, but significant, sign of the aforementioned might be found in Young's analysis approach of light, wherewith drawing consistent analogies from sound and anatomy-based acoustics. Ultimately, it led Young to scientifically acculturate himself with harmonic and music theory – Bernoulli, d'Alembert, and Euler – and to further extend his original metaphor to experiments on vibrating organ pipes and refraction, reflection, and inflection (diffraction) on thin colour plates. This is important, as it severely contrasts, at least with the Kantian dualist (thought *vs.* language)-Three Critiques period interpretation of language in the guise of *in concreto* subjective empirical dwellings, linearly reflected in the passage:

"To one man, for instance, a certain word suggest one thing, to another some other thing; the unity of consciousness in that which is empirical is not,

as regards what is given, necessarily and universally valid" [2] (*Critique of Pure Reason*; B 140)

which shows how the philosopher from Königsberg understood language (again, leastwise in the dualist-Three Critiques period), as being a sort of inertial, at rest or moving frictional (anatomical) body in a symbolic metaphysically nullified straight line, unless acted upon by a force, one such as the power of the concepts of reason. Most obviously, nowhere close was therein present one *Aufklärung* interpretation in assent with a pervading wave-theoretical mediumist interference.

Thereafter, it was Young's interpretation of resonant vibrating reed pipes in analogy with thin plate colours, which eventually permitted the shift of understanding from cavity peak pulses harmonics, to the much broader understanding of coalescence of "undulations", a term that was eventually ratified, against (Hellenistic) "emission" or (Cartesian) "vibrations", in his second Bakerian lecture (12 November, 1801), also the place and time of the physiological three-receptor perception of colour theory, precisely on the basis of anti-corpuscular stigma, even though one such concept as that of "interference" was absent:

"When two Undulations, from different Origins, coincide either perfectly or very nearly in Direction, their joint effect is a combination of the Motions belonging to each." [3] (Thomas Young; Proposition VIII; 1802b *On the theory of light and colours*. PT, 92 [read 12 November 1801].)

Polarization, in turn, came upon once it was observed by the French physicist Malus, during the course of experiments on double refraction, with light being reflected from the windows of the Luxembourg palace through a Iceland crystal, similar to Huygen's verification of double refraction on calcite, and prior to these Newton's. Yet, it was only with Malus that polarization was seen as independent of crystalline media. Suitably, reflection was only ordinary when the crystal's plane of reflection was parallel to its axis. The following work of the only remnant Newtonian, and Scottish scientist, David Brewster (1781-1868) on the polarizing angle of crystals, besides other crucial experiments with different apparatus, permitted the postliminary work of Arago (1786-1853) and Biot (1774-1862) on chromatic polarization, detaching it from Laplace and Newton, and even from Fresnel's hypothesis of the molecular ether, thus more and more towards a full-blown wave-theory of light based on purely transverse waves.

Anyhow, polarization as transverse waves which externalize the geometrical orientation of the oscillations, flagrantly epitomizes a *kybernetikos* sense, originated from the Leibnizian *s'apercevoir*, which Kant rescued by including "apperception" in his philosophical vocabulary, thus inextricably reciprocate with *Denken orientiren* (1786), about which Young's double slit experiment is certainly transmutable with Kantian *antinomien* ("antinomies"), in the sense that a single wave-front, herein understood as the theoretical concept of pure reason, is coherently split off and made orthogonally interfered.

The epilogue to this was the discovery of the photoelectric

effect by Albert Einstein (1879-1955) in 1905, thus the longer axis of the elliptic serendipity that we alluded to in reference to *Enlightenment*, the final paradox being that the Newtonian "corpuscular" study of polarity as one only slit light plane would lay the groundwork for the wave-theory of light, which eventually would culminate in Bohr's (1885-1962) model of the atom, yet under an utterly statistical wave-particle duality. With all these elements taken *sub judice*, our ground thesis is that the period at stake (1666-1801) positively vindicated the Newtonian-to-Kantian Enlightened corpuscular thing-in-itself (*noumenon*; *ding an sich*). Hence, in a *kybernetikos* viewpoint, it seems it was more about an observing faith in pure reason, rather than a critique of pure reason.

A better understanding of Enlightenment is achieved, albeit in backlight, by means of what the Newtonian-to-Kantian Enlightened corpuscular thing-in-itself (*noumenon*; *ding an sich*) stood against: natural-empiricist methodological skepticism, or more strictly, the Humean "new scene of thought" [4] (*HL* 3.2). David Hume (1711-1776) naturally acquiesces and overlooks the philosophical modernity, being late to the disputation and quarrels between early British empiricism – Francis Bacon (1561-1626) and Thomas Hobbes (1588-1679) –, dualistic and monistic rationalism – respectively, Descartes (1596-1650), and Spinoza (1632-1677) –, fideism – Pascal (1623-1662) – and, in the end, monadological rationalism and empiricist idealism – by order, Leibniz (1646-1716) and Berkeley (1685-1735) –, which conforms to a sort of *Der Wanderer über dem Nebelmeer* (Wanderer above the Sea of Fog; oil painting c. 1818) Nietzschean firm stance over Romanticism (Goethe, Hölderlin, Wagner), in Hume's case not exactly of nihilism, but instead a powerfully inductive common-reason deep stand for skeptical subjectivity.

It must have been along these lines of metaphysical necessary *a priori* contingency, that the American philosopher David Lewis (1941-2001) laid ground for the principle of "Humean supervenience" [5], on the basis of a metaphysical mosaicizing. By all means, in the context of a deviated critique warning call, and a *kybernetikos* full reappraisal, it urges to cut across important aspects of Hume's philosophy.

Accordingly, and in spite of Hume's dramatic reception of Joseph Butler's *The Analogy of Religion: Natural and Revealed to the Constitution and Course of Nature* (1736) – whose apologetics were sketching signs of a pre-Kantian transcendental ideal (*prototypon transcendentale*) in relation with an objective idea of practical reason, favouring a deistic design of nature, in detriment of the late Kantian's notion of conformity to a natural end (*zweckmässigkeit*) withstanding a prefiguration of Darwin's natural selection if asserted on purely critical empiricist phenomena, the philosophical masterpiece *A Treatise of Human Nature* (1739–40) had already been published and made known to the world, which and in turn, as it is proverbially familiar, would awake Kant from his "dogmatic slumber" [6].

Less trivially, though, it stands out the unperturbed critical frontiers set out from Humean philosophy, as the summit of Scottish Enlightenment, which Kantian critical philosophy

never resolved. It deserves notice that, therein, the entire universe of primary and secondary qualities philosophically undergoes a *synthesis*, simultaneously a collapse of both space & time, as of logical causation. This is all echoed in one work – both under *analytical* and *dialectic* impressionist (ideas being faint “images of impressions” [7] (*THU* 1.1.1.1/1) Book I “Of the Understanding”, and Book II “Of the Passions” (1739), and later in the shortened covering of the same topics in the *Enquiry* (1748) – with a clear “anatomy of the mind” [7], [8] (*EHU* 1.13/13; *THU* 2.1.12.2/326) anthropological orientation, therefore naturalistic empiricism bringing Newtonian *Principia* to mere perceptions.

The search for the “proper province of human reason” [8] (*EHU* 1.12/12) suddenly meets Newton’s proper scientific method, under the determining norm to derive “all effects from the simplest and fewest causes” [7] (*THU* 7.8). But, beyond these facts behind the promotion of the Kantian Copernican revolution, under a much broader sense of Bentham’s “fallen scales” [9], what is exceptional is that Humean’s guiding principle of association – “resemblance, contiguity in time and place, and causation” [8] (*EHU* 3.2/24) – rests upon “a gentle force, which commonly prevails” [7] (*THN* 1.1.4.1/10), of which gravity is just the frontispiece established principle, a sort of common ground derivability *synthesis* between the *a priori* and *a posteriori* realms, or *as if* – (*als ob*) in Kantian’s recognition – the gravitational law principle of association, as a “constant conjunction” [7] (*THN* 1.3.6/93) between impressions and universal ideas, was the closest ground of an idea to a perception. Moreover, it was also the very same nature of the principles of association present in the *Dialogues Concerning Natural Religion* (1779) which offered the very first commonly prevailing naturalistic interpretation that guided Charles Darwin’s work, over Kantian’s *Critique of Judgment* (1790). Thereby, it is fully realised how Hume’s account of the power of definition was, in his words and very suitably for the appraisal of the question – *Was ist Aufklärung?* (Kant, 1784) –, a “new species of optics” [7] (*EHU* 7.1.4/62). And, thus, our thesis, taking French physicist André-Marie Ampère (1775-1836) *Essai sur la philosophie des sciences* (1834) “Cybernetique” [10], is that under the truest optic and philosophical oriented sense, *Enlightenment* corresponded to the strongly empiricist and inductive critically scientific use of reason, as the tenet or demonstration and close or comeback, of Scottish Enlightenment, from Hume to Maxwell.

III. OPUSCULES ON THE SEVEN REMARKS TOWARDS CRITICAL TRANSCENDENTAL EMPIRICISM

Hereby we methodically expand them in concise form, by way of summary and further research program, with the corresponding indentation:

1) Darwin never used “evolution” in any of his works, a concept which carries severe misconceptions in relation to the metaphysical arrow of time, but instead “natural selection” [11] or “descent with modification” [11]. Kant’s *Critique of Judgment* (1790) makes innovative use of adequation

(*Zweckmäßigkeit*) in relation to nature, by means of reflective judgment (*reflektierende Urteilskraft*), although dismissing it, in favour of the final end of reason (*Vernunftendzweck*), whereas from Darwin, biological and system science sorted out from the previous the regularity law (*Gesetzmäßigkeit*) of “natural selection” [11] and “adaptation” [11] as self-organized circular causality purposeful nature (*Natur*), Humean in essence.

If the power of *synthesis* as “the action of putting different representations together with each other and comprehending their manifoldness in one cognition” [2] (*CPR*; A77/B103) is clear under the “modern synthesis of evolution” [12] (Julian Huxley, 1942), the 21st century has seen modern synthesis in biology logical extension products (molecular, microbiological, and genomic phylogeny extensions, but also embryological development, morphology, and ecology ontogeny extensions).

Yet, presumed therein is always, more so and statistically, a scientific timely “synthesis”, never a (natural sciences) dialectic “synthesis”. A biological and life sciences “Analytic of Principles” [2] is, thus, beyond the liaison between causality and phenomena (hence an idealist misleading transcendental schema) critically bounded to a conjunction of all representations, which means that it shall oscillate between analogies of experience and postulates of empirical thought. Dialectic complexity as “the employment of our faculty of cognition from principles, and with it philosophy, is coextensive with the applicability of *a priori* concepts” [2] (Kant, *Critique of Judgment*), which is the same as saying that what is evolving is the “synthesis” itself. The Kantian vitalised concept of “spontaneity” [2] (*Gemüt*), as self-consciousness of the total unity of possible knowledge, a very similar concept with the pure emergence of perceived time, surely demeans every rationalist prospect, although not the science of biology. But, more interestingly, the self-organizing life concept, or Aristotelian-influenced “*matière à pensée*” [13] (Changeaux, 1988), under the concept of “order-from-disorder” [14] (Schrödinger, 1944) is, surely, the major original *kybernetikos* entropic direction of either irreversible or possible phenomena. On the whole and at the very least, regarding the life sciences, we do know what does it mean to orientate in thinking, for the sole reason that natural sciences phenomena are, themselves, inasmuch critical as statistical, transcendental empiricist beings and objects, i.e., a metaphysical naturalism.

2) Maxwell’s electromagnetism mathematical formalism on Faraday’s electromagnetic induction was 19th century radiation and wave-like “synthesis” (*Verbindung*) (magnetism, electricity, light), second to Newton’s unification. Although developed from mechanics (vector and tensor analysis, potential theory, calculus partial differential analysis) and Enlightenment mathematicians’ work on the wave-equation (d’Alembert, Euler, Bernoulli, and Lagrange), it endowed new “Metaphysical Foundations of Natural Science” (Kant, 1786) based on a field of distribution of static point charges,

polarization displacement, self- and mutual induction, energy flow and free space, with a strong logical and scientific re-steering comeback of Scottish Enlightenment, at the same time "kybernetes" first exhibition (*Darstellung*) with *A Dynamical Theory of the Electromagnetic Field* (1865).

Often unnoticed, but symptomatic, is the effort that neo-Kantianism invested, through the leading and tentative work of Herman Cohen (1842-1919) at Marburg University, while promoting the transcendental method, of reacting to Faraday's "doctrine of electricity" (*Elektrizitätslehre*), similar to the German Jewish sketched philosophical response in the *Principle of the Infinitesimal Method* (1833). Both Frege (1848-1925) and Russell's (1872-1970) criticism of Cohen are the best possible evidence to understand how the *calculus* bearing on infinitesimals, rather than limits, lodged great danger to the Kantian idea of metaphysics, and loomed severe peril to the *Aufklärung's* idea of pure mathematics. Again, in an elliptic serendipity, it was not only Russell's study of Frege's logicist program that entailed the so-called "Russell's paradox", inasmuch as the overlooked Cantorian "diagonalisation" was tooled by Kurt Gödel (1906-1978) to self-inflict the Platonist realm of mathematics with the paradoxes of the infinitesimal and the continuum, by the incompleteness theorems (1931). In regard to Cohen on Faraday's work, it is pivotal to grasp how, besides remarking about it the "direction of which modern science tries to orient itself" [15] (*Introduction to Friedrich Lange's History of Materialism* 1877), it is revealed, against a Newtonian-like force, an energeticist theory (*Energielehre*) that mirrored "the fundamental tension between atomism and dynamism" [15] (Lydia Patton, 2004). Boltzmann's (1844-1906) image (*bilder*)-interfering atomism, against an ethereal and "hidden medium" (Maxwell and Mach), was, thereafter, to be developed from the inherently statistical kinetic theory of gases, which forms another manifestation of an elliptic serendipity, as if (*als ob*), Kantian antinomies (*antinomien*) had met Cantorian diagonalisation (*Diagonalargument*) in "fundamental tension" [15].

3) Young's double-slit experiment (1804) was the first light and matter double-path experiment, contrary to the Enlightened similar in cosmology, but massless and unobservable (mathematical and dynamical) Kantian antinomy of pure reason, all in all a Kantian construct subdued from Newton's noumenal and corpuscular light theory, truly a *natural minima* metaphysics with respect to light and its shadows (scholastic), orientated by an eminently moral practical (*praktischen*) judicial reason. Yet, prior to the ground of cosmology, the "Appendix on the Amphiboly of the Concepts of Reflection" [2] (*CPR*, B316) preparing the passage to "Transcendental Illusion" [2] in "Transcendental Dialectic" [2] (*CPR*, A293) and, namely, paralogisms at the heart of psychology, along the previous "Postulates of Empirical Thought (Refutation of Idealism)" [2] (*CRP*, B266-B275) herein meaning "material idealism" and, thus, strong skepticism (Hume, Berkeley, Descartes), are unequivocal

steering (cybernetic) errors in relation to the "Analytic of Principles" [2], to which Young's double-slit experiment was, specially considering late Kantian architectonic keystone work – *Religion Within the Bounds of Bare Reason* (1793), *Perpetual Peace* (1795), and *Metaphysics of Morals* (1795) –, the fittest feedback control (as to axioms of intuition, anticipations of perception, analogies of experience, and empirical postulates of thought) in a proper enlightened medium. Ostensibly, it was set to reassure a non-dualistic perceptive over dualistic representational nature.

It is also important to recognize that Young, whose own biography bridged Edinburgh of Hume to Göttingen of Cantor, established, through his original two-pinhole light interference experiment and, consequently, its modern variants, a preponderant critical transcendental empiricism.

If similar to the coherent light source in the original experiment, we could devise a coherent philosophical principle, even ahead of physics Bohr's complementarity principle (1927) and Heisenberg's uncertainty principle (1927), it prevails the sort of synthetic *a posteriori* principle in the power of association and in the power of association only. Only through the principle of association – "resemblance, contiguity in time and place, and causation" by "points of contact" [7] (*THN* 1.4), commented by Hume as the most noteworthy of his discoveries and the "cement of the universe" [16] (Abstract 1740) – is it possible to confer a philosophical heuristics to the wave-particle duality probabilistic interference, even if tormented with such terrible theoretical conundrums of probability-derived logical collapse, two-paths superposition, or "annihilation of time" [7] (*THN* 1.3).

4-7) Cantor's diagonal argument (1891) goes surprisingly unnoticed as the dominant new method and proof technique, at the root of contemporary metaphysics and philosophy of science, having cast light to Gödel's incompleteness theorems (1931), a violent blow to synthetic *a priori* judgments and mathematics, the negative answer to the *Entscheidungsproblem* (Church-Turing, 1936) and, thus, to effectively calculable functions (or computable functions) in a Turing-machine, with corollaries in mathematical information theory (Shannon, 1948), having even incremented both theoretical and practical prolegomena of physics, such as Bohr's complementarity principle measurement problem (1927), and mathematical inequality Heisenberg's uncertainty principle (1927), to which we could add countable bijective correspondences and denumerable structures in evolution (DNA structure, 1962). Cantor's diagonal argument (*kybernetes*) quivers and oscillates Kant's antinomies (*antinomien*), dismissing "Dialectic Inferences of Pure Reason" [2] (*CRP*, A293) with "non-empty" pure reason as the axiom of choice, and analytically re-centers the "Seat of Transcendental Illusion" [2] (*CRP*, B355), not only by means of different orders of infinity, non-denumerable power sets of Reals (*a priori* mathematics), towards superinduction and arbitrary well-ordering, but also grants the proper displacement of

the continuum (Gödel, 1931), and formal logic (strictly non-transcendental) upturn with the undefinability of truth (Tarski, 1933), as a sort of momentum serendipity with the Copenhagen interpretation in physics (Bohr & Heisenberg, 1927).

The greatest example of one such metaphysical diffraction, or rather said a paradoxical *Enlightened* refraction through the medium of pure reason, peremptorily diagonalised by Cantor, was the work of Grete Hermann (1901-1984). Indeed, the German philosopher and mathematician pioneering work on the theory of polynomials (1926), and critic of von Neumann's (quantum) hidden-variables theorem (1935) is unachievable without a deep intuition of diagonalisation. Grete Hermann, as a disciple of the Neo-Kantian Friesian School Leonard Nelson (1882-1927), authored the book *The Natural-Philosophical Foundations of Quantum Mechanics* [17] (1935), a result of her visiting Heisenberg and Weizsäcker in Leipzig for the sole purpose of establishing a Kantian synthesis with quantum mechanics, a result of which being also Heisenberg's chapter "Quantum Mechanics and Kantian Philosophy" in *Physics and Beyond* [18] (1969). Nevertheless, even if contemporizing with Heisenberg's much later appreciation of Kantian philosophy compared with quantum mechanics, and taking into account not only the way Grete Hermann diverged from an hidden variables theory, inasmuch as Heisenberg diverted from a thing-in-itself (*noumenon*; *ding an sich*) interpretation unless framed into a nomothetic heuristics, it seems to us that, yet again, a steering *kybernetikos* error was produced. The "condition of possibility" [2] (*Bedingungen der Möglichkeit*) of knowledge that is being critically trialed under critical transcendental law in one such tribunal of reason as that of Young's original interference experiment and its variants, is on no account universal and necessary, nor is it decidable or complete, reversible or predictable, unless diagonalised, statistically computable (Turing-Church, 1936) and informational (Shannon, 1948), and where the basis on which the transcendental method lies must also assume to be critically bounded to formal logic (Tarski, 1935).

In actual fact, the Hermann & Heisenberg neo-Kantian interpretation of quantum mechanics, even if the mathematical and physical study on visibility and distinguishability was scientifically impeccable, at the end was finally found to diffract to the extreme, or rather said to reflect, and not without contradiction, the absolute idealism (*das Absolute*) of Hegel, away from Fries's trilemma. Verifiably, about the core of Hermann's theory in relation to local causality (position) and time (momentum) neo-Kantian re-validating assertions, Léna Soler wrote:

"The results of measurements actually obtained for quantum objects cannot be univocally predicted with certainty. However, *after* having performed a quantum measurement, and *after* having gained knowledge of its result (previously not predictable with certainty), it is possible, by working backwards, to reconstitute, retrospectively and completely, the

causal chain which has necessarily produced such a result." [17]

IV. CONCLUSION

It is our thesis and work program that, under a *kybernetikos* sense of orientation, the philosophy of David Hume as the epitome of Scottish Enlightenment answers, for any philosophy of science in time, the question "What Does It Mean to Orient Oneself in Thinking?" (Kant, 1786). This does not elude the necessary "synthesis" with Kantian critical philosophy, namely the transcendental method as a common ground to Neo-Kantianism, and even critical philosophy.

As for the latter, the birth and spectacular growth of cybernetics in the late 1940s (Wiener, McCulloch, von Neumann, up to Maturana and Varela in the 1970s), agrees, in historical terms, with the most consensual modern breadth in critical theory (Horkheimer, Adorno, Marcuse, Habermas). However, it is mainly by recognizing cybernetics as "The science and art of understanding of understanding" (Rodney Donaldson), why the ground of Kantian metaphysics is visibly appurtenant. The all-abridging, very informal, Macy Conferences [19] (1941-60) took part in the spring of cybernetics by a series of conferences (1946-53), during which course the concept was coined by Norbert Wiener. Out of this group of altogether ten conferences there are unfading topics of research. However, to the assumption that a proper philosophical oracle to this interdisciplinary force was absent – and beyond control systems and computer science, evolutionary biology and complexity sciences, neurosciences and psychology –, with this paper we propose a cybernetic philosophical research within the bounds of critical transcendental empiricism, across the grounds of pure common reason.

The construed meaning of architectonics in Kant represents, as was remarked by Paul Guyer [20], a departure from the Vitruvian "*ordinatio*" and "*dispositio*", "*eurythmia*" and "*symmetria*" alike gamut of concepts, perennial until Christian Wolff (1679-1754), which, in addition with intermingled elements from Baumgarten (1714-1762) and Lambert (1728-1777), sustained a metaphysical edifice that was already dissimilar under the next generation of philosophers (Hegel, Schelling, Schopenhauer). Even so, Kant's architectonics might be misleading, once he calls it "the art of constructing a system" (CPR A832/B860), going as far as comparing it with an inner-folding growing organic *articulatio per intussusceptionem* animal body, which ultimately ought to align with the "physiology of pure reason" (CPR A845/B873) and the biogenetic purposiveness presumed in the developmental concept (*entwicklungsgeschichte*) (cf. Adrian Switzer, 2014) [21]. So near and yet so far, Kantian architectonics is, though, oriented beyond Baumgarten's four-part traced *Metaphysica* (1739), towards the "original idea of a philosophy of pure reason" (CRP, A847/B875), namely the speculative (nature) and practical (freedom) use of pure reason (CRP A841/B869), the latter of which would constitute in the form of apodictic law, or pure practical reason, the "keystone" (*Schlußstein*) of the system as a "whole" (*ganze*). Hence, it cannot be disguised

the fact that the advocated values of a complete "scientific" (*szientifischen*) or "systematic" (*systematisch*) metaphysics – in truth, as a whole neither complete nor consistent at the root of mathematics, as enlightened by Cantor and Gödel, and without any provision in nature or freedom as proclaimed by Hume, neither of the two redeemable by dialectic form – are a *kybernetikos* critical misalignment in relation to the proper transcendental philosophy.

The "Transcendental Analytic" output of possibility of objectivity in general succumbs to "transcendental illusion" (*transzendentalen schein*) "paralogisms", "antinomies", and, ultimately, to the "ideal" of pure reason. Contradictorily, it is, at last, on the basis of the "connecting together" (*zusammenhagen*) free interplay affiliate with nature, that Kant envisaged the root of metaphysics as the science of the first principles of the entirety ("conexium"-to-"ens" dialectic, i.e., as true as apparent, in bounds and out of the bonds, limiting and overextending) of human cognition. With this, nature, really, ought to "was sein soll" (CRP, A845/B873) become just the rational *theoria* or contemplation of nature ("der Grebauch der Vernunft in dieser rationalen Naturbetrachtung") (CRP, A845/B873), never nature as the whole as it exists (*alles, so fern es ist*) (CRP, A845/B873). This *per se* justifies our enquiry about an inversion of the culturally distinguished Oriental Prussian Enlightenment Kantian architectonics.

Not even a sort of transcendental entangled superposition of analytical (mathematical) and dialectic (dynamic) states of nature and freedom, under the Heisenberg & Hermann quantum interpretation, nor a Gödelian (Platonic) *summum bonum* belief could ever rebirth "interconnection" (*zusammenhang*) under "pure reason" (*der reinen Vernunft*), in favour of the "critical path" (*kritische weg*).

This is important, insofar systems design ontology is the contemporary "technical" (Kant, CRP, A847/B875) version of architectonics towards the constitution of sciences, and to a greater degree if a new Enlightenment age is to be enacted by cybernetics. There is, surely, following the high disturbance physicality of light – yet never so herein "transcendental" understood as non-immanent "hyperphysicality" (CRP, A846/B876) – a sense where the constitution of sciences is cognate with the varying (electromagnetic) fields, which are always perpendicular to the direction of motion of the light, i.e., rather "diagonal" (Cantor) or mutually rectangular, than "antinomic" (Kant) or parallel equidistant. But what we wish to call attention to, summarizing, is the critical observable constitution of sciences, as of philosophy of science vis-a-vis the polarized *kybernetikos* orientated nature of light:

"(...) if we look at a ray of light head on as it comes toward us, the electric field may rotate in a circle, either clockwise or counterclockwise (circularly polarized light); or it may trace out an ellipse, rotating and varying in length; or it may simply oscillate back and forth without rotating, always remaining in a plane that points in a given direction." [22]

Not only the goal is to clarify the intended meaning of an

"elliptic serendipity" in philosophy of science directed toward a sound theory of scientific knowledge, as it is also made possible to throw light on the possibility of a *kybernetikos* orientated system design critic transcendental "ontology", irreducible to metaphysics, as "the system of all the conceptions and principles belonging to the understanding and the reason, and which relate to objects in general, but not to any particular given objects" [23] (CRP, A845/B873).

REFERENCES

- [1] A. T. Nuyen, "Kant on Miracles," *History of Philosophy Quarterly*, vol. 19, n°43, pp. 309–323, July 2002.
- [2] Immanuel Kant, *Critique of Pure Reason*, Norman Kemp Smith, Published in Great Britain, R. & R. Clark, Limited, Edinburgh.
- [3] Thomas Young, "Miscellaneous Works", vol. 1: 157, in Thomas Young, the last man who knew everything, Andre Robinson, One World, Oxford, 2006, p. 106.
- [4] The Letters of David Hume, edited by J.Y.T. Greig, 2 volumes, Oxford: Clarendon Press, 1932.
- [5] David Lewis, *On the Plurality of Worlds*, edited by Basil Blackwell, 1986, pp. 14-16.
- [6] Immanuel Kant, *Prolegomena to Any Future Metaphysics with Selections from the Critique of Pure Reason*, Edited by Gary Haltfield, (Revised Edition) Cambridge Texts in the History of Philosophy, Cambridge University Press, 2004, p. 10.
- [7] David Hume, *A Treatise of Human Nature*, edited by L. A. Selby-Bigge, 2nd ed. revised by P. H. Nidditch, Oxford: Clarendon Press, 1975.
- [8] David Hume, *An Enquiry concerning Human Understanding*, edited by Tom L. Beauchamp, Oxford/New York: Oxford University Press, 1999.
- [9] Jeremy Bentham, (1776) *A Fragment on Government*, Cambridge Texts in the History of Political Thought, Cambridge University Press, 1988, pp. 36-60.
- [10] André-Marie Ampère, *Essai sur la philosophie des sciences, ou, Exposition analytique d'une classification naturelle de toutes les connaissances humaines*, Bachelier, Paris, 1834.
- [11] Charles Darwin, *The Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life*, Cambridge University Press, 2010.
- [12] Julian Huxley, *Evolution: the Modern Synthesis*, Allen Unwin, London, 1942.
- [13] Jean-Pierre Changeux et Alain Connes, *Matière à pensée*. Odile Jacob, Paris, 1989.
- [14] Erwin Schrödinger, *What is Life?: With Mind and Matter and Autobiographical Sketches* (Canto Classics), Cambridge University Press, 2012, pp. 3-19.
- [15] Lydia Patton, Hermann Cohen's History and Philosophy of Science, Library and Archives Canada, Published Heritage Branch, "The Einleitung mit kritischem Nachtrag zu Friedrich Albert Langes Geschichte des Materialismus" (244-5). 2004. (selected translation), p. 125. [1914] in the bibliography, hereafter cited as *Einleitung* 244-5
- [16] David Hume, *An abstract of a book lately published; entitled A Treatise of Human Nature, & c. Wherein the chief argument of that book is farther illustrated and explained*, Eighteenth Century Collections (ECCO), London, 1740, p. 36.
- [17] Grete Hermann, *Natural-Philosophical Foundations of Quantum Mechanics*, Elise Crull, Guido Bacciagaluppi (Editors), *Studies in History and Philosophy of Science*, Springer, Volume 42, 2016.
- [18] Werner Heisenberg, *Physics and Beyond, Encounters and Conversations*, Edited by Ruth Nanda Ashen, World Perspectives, Volume Forty-two, Harper & Row, 1971.
- [19] Cybernetics, *The Macy Conferences 1946-1953, The Complete Transactions*, Edited by Claus Pias and Joseph Vogl, Diaphanes AG, 2016.
- [20] Paul Guyer, "Kant and the Philosophy of Architecture", *The Journal of Aesthetics and Art Criticism*, Volume 69, Issue 1, 2011, p. 7-19.
- [21] Adrian Switzer, *The Traditional Form of a Complete Science: Baumgarten's Metaphysica in Kant's Architectonic of Pure Reason*, *Philosophica*, vol. 44, 2014, pp. 149-164.
- [22] Tim Maudlin, *Quantum Non-Locality & Relativity, Metaphysical Intimations of Modern Physics*, Wiley-Blackwell, 2011, pp. 8-9.
- [23] Immanuel Kant, *Kant's Critiques*, Wilder Publications, 2008, p.379.